Results

Results I

Run snapshot. With the bundled `manuscript/config.yaml` the most recent execution evaluated the query *"reproducible research optimization"* against local, returned 6 deduplicated paper(s) (4 carrying a DOI, 6 carrying an abstract); the per-source breakdown is local=6 and recorded backend errors are none. The deep-search workflow ([@sec:deep_search]) covered 3 keyword(s) — *convex optimization; stochastic gradient descent; reproducible research* — drawn from arxiv, crossref, producing 300 unique paper(s) after cross-keyword deduplication.

When `sources: [local]` is used, this section reports fixture execution only. It must not be read as a claim about empirical literature coverage; a live-provider run requires source-level provenance and a separately reviewed claim boundary.

Results II

The diagnostic figures generated for this run are catalogued in [@sec:methodology]: [@fig:papers_per_source] surfaces per-backend coverage, [@fig:year_histogram] surfaces the temporal distribution, and [@fig:score_distribution] surfaces the relevance-score profile. The full determinism contract for each stage is itemised in [@tbl:determinism] of [@sec:reproducibility].

Interpreting the run snapshot I

The numerical values in the run-snapshot paragraph that opens this section are read directly from `output/run_summary.json` and `output/data/manuscript_variables.json` so they always reflect the most recent run rather than a stale claim hand-typed into the prose. Three properties are worth highlighting (the formal determinism contract for each underlying stage is enumerated in [@tbl:determinism]):

- **Cache reuse is observable.** A second invocation against the same config produces a byte-identical artifact tree (modulo the wall-clock timestamp inside `output/search/cache/search_<hash>.json` itself); see also the *Search (cached hit)* row of [@tbl:determinism] and the verification recipe in [@sec:reproducibility]. The cache file thus doubles as a cryptographic seal: re-running is a file read, not a network round-trip.

Interpreting the run snapshot II

- **Deduplication is signal-preserving.** The aggregator merges by DOI → arXiv id → normalised (title, year), keeping the highest-scored copy and filling missing fields from the loser (see *Dedup / merge* in [@tbl:determinism] and the per-backend pre-dedup view in [@fig:papers_per_source]). The `RESULT_NUM_PAPERS` figure therefore equals "papers a reviewer needs to read", not "raw backend hit count" — the per-source contributions in `RESULT_PER_SOURCE` are the pre-dedup view.
- **Enrichment coverage is honest.** `RESULT_WITH_ABSTRACT` and `RESULT_WITH_DOI` count fields the corpus or the `AbstractFetcher` actually populated, never values inferred. When a paper is missing a DOI it is excluded from the DOI count even if its arXiv id resolves to one upstream. The temporal coverage of those papers is summarised by [@fig:year_histogram], and their backend-reported relevance scores by [@fig:score_distribution].

Output artefacts I

After running `scripts/run_search_pipeline.py` against the default `manuscript/config.yaml`, the project produces:

- `output/search/results.json` — the raw `SearchResult` JSON, including `per_source_counts` and `errors` for diagnostic purposes.
- `output/search/cache/search_<hash>.json` — the deterministic search cache; identical reruns are file reads.
- `output/cache/abs/<safe_id>.txt` — one file per fetched abstract.
- `output/cache/pdf/<safe_id>.{pdf,txt}` — PDFs and extracted text (only when `enrichment.fetch_fulltext: true`).
- `output/corpus.json` — a `LocalBackend`-compatible JSON corpus of every result, enriched in place.

Output artefacts II

- `manuscript/references.bib` — the auto-populated bibliography from the single-query pipeline (merged with any other `manuscript/*.bib` at PDF render time).
- `output/llm/per_paper/<safe_id>.md` — per-paper LLM analyses (only when `llm.per_paper: true` *and* the LLM stack is reachable).
- `output/llm/synthesis.md` — corpus-level LLM synthesis (only when `llm.corpus_synthesis: true` *and* the LLM stack is reachable).
- `output/reading_report.md` — the final assembled reading report.

When the LLM stack is genuinely unreachable, the `output/llm/` artefacts are simply absent — no placeholder file is ever written into the archive (see [@sec:pipeline_internals]).

Output artefacts III

Because the search cache and abstract cache are deterministic, a second run with identical `config.yaml` produces byte-identical artifacts (modulo timestamp metadata in the cache files themselves). This is the property the project exists to demonstrate.

The exact paper count, DOI list, and synthesis text depend on the live state of arXiv and Crossref at the time of the run — and are therefore not reproducible *across* runs in different weeks. Users seeking strict reproducibility should:

1. Pin a `LocalBackend` corpus generated from a successful run (`infrastructure.search.literature.write_corpus`) and remove `arxiv` / `crossref` from `project_config.search.sources`.
2. Commit the `output/search/cache/` directory to version control.
3. Pin the LLM seed (`config.llm.seed`) and avoid model upgrades.

Output artefacts IV

With those three steps, every run from the same commit produces the same outputs.